SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMKUR

(A constituent college of Sri Siddhartha Academy of Higher Education, Tumakuru)

Department of Electronics and Communication Engineering

22EC303: Digital Circuit Design Using Verilog

Date:07/11/2023 **TEST-1** Time:1.00Hr

Answer all the questions

		M	C	B
1	Differentiate minterm and maxterm.Express the following Boolean function in m-notation. i) (a+b) (a+c) ii) ab' + cb +ca'	6	1	3
2	Simplify the following using K-map. i) f(w,x,y,z) = ∑ m (0,2,4,5,6,8,10,15) + ∑d(7,13,14) ii) f(a,b,c,d) = πM (0,4,5,7,8,9,11,12,13,15).	6	1	3
3	Write a Verilog code for the full adder using data flow and behavioral modeling.	6	1	1
4	Design a minimal sum combinational circuit to generate Odd parity for 8421 BCD Code.	6	1	3
5	Define decoder. Explain 2:4 decoder with active high enable input.	6	2	1

NOTE: M is marks, C is CO and B is Blooms level

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Date:12/12/2023 **TEST-2** Time:1.00Hr

Answer all the questions

		M	C	B
1	What is comparator? Design one bit comparator.	6	2	1
2	Summarize the function of BCD adder with neat block diagram	6	3	1
3	Design 4:1 multiplexer and write a Verilog code using gate level modeling	6	4	3
4	Explain with a neat block diagram and waveforms describe the application of SR latch as switch debouncer.	6	2	1
5	Implement the following Boolean function using 4:16 decoder with active low type. i) $F_1(W,X,Y,Z) = \sum m(2,4,6,10,14,15)$ ii) $F2(A,B,C,D) = \pi M(2,3,10,11,12)$ iii)Differentiate between decoder and encoder	6	1	3
	NOTE: M is marks, C is CO and B is Blooms level			

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22EC303:Digital Circuit Design Using Verilog

Max. Marks:20 TEST 3 Time:1.00Hr

Date : 02-01-2024

Answer all the questions

		M	C	B
1	Write the logic diagram, truth table and derive the characteristic equation of i) D flip flop ii) JK flip flop	5	2	2
2	Mentioning any one application of shift register, explain 4 bit Serial input and parallel output(SIPO) shift register.	5	3	3
3	Write the verilog code of 4 bit BCD UP counter in behavioral style	5	4	2
4	Design 2-bit asynchronous UP counter. Explain the same with neat waveforms.	5	2	4

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SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY- TUMAKURU

(A constituent College of Siddhartha Academy of Higher Education, Tumakuru)

B.E., SEMESTER END EXAMINATION JAN - FEB 2024

22EC303 : DIGITAL CIRCUIT DESIGN USING VERILOG

TIME: 3.00 Hrs **SEM: III** **MAX MARKS: 100**

NOTE: Answer any five full questions selecting one full question from each choice.

		M	CO	BL
1.a)	Express the following using m/M – notation. F1= ab + bc' +abc' F2= (x+y) (y'+z') (x' + z')	6	1	2
b)	Simplify the following using K-map i) F(a,b,c,d) = ∑ m (1,3,7,11,15) +∑ d (0,2,5) ii) F(w,x,y,z) = πM (0,4,6,8,9) + d(1,12,13,14,15)	8	1	3
c)	Describe the structure /components of a Verilog Module with an example.	6	4	1
	OR			
2.a)	Define the terms i) Minterm ii) Literal iii) Maxterm iv) Canonical Form	6	1	1
b)	Design a minimal sum combinational logic circuit to generate odd parity for 8421 BCD code.	8	1	3
c)	With the neat flow chart describe the typical design flow of VLSI IC circuit design.	6	4	2
3.a)	Design 1 – bit comparator.	6	2	3
b)	Signify the select lines in multiplexer. Implement the following boolean function using i) 8:1 Multiplexer ii) 4:1 Multiplexer, consider LSBs for select lines. F(A,B,C) = Σ m (2,4,5,6)	8	2	3
c)	Write a verilog code for the half adder using data flow and structural modeling	6	4	2
	OR			
4.a)	Justify the statement "Encoder is a combinational Circuit". Write and explain the IC diagram of 8:3 priority encoder.	6	1	1
b)	Signify the advantages of carry look ahead adder over parallel adder with necessary equations and logic circuit.	7	3	2
c)	Write the logic circuit of 4:1 Mux and develop the Verilog code for the same using structural modeling.	7	4	3
5.a)	Discuss the significant charecteristics of basic bistable element with neat logic circuit.	6	3	2
b)	What is characteristic equation? Write the truth table and derive the characteristic equations of the following flip flops i) JK Flip flop ii) D Flip flop	8	2	1
c)	Illustrate the significance of blocking and non blocking assignment statement in behavioural modelling.	6	4	3
	OR			
6.a)	Mentioning the truth table, summarize the working of SR latch constructed with NOR gates.	7	2	2
b)	What is race around condition? Interpret how it is overcome in MS JK flip flop with neat logic circuit and waveforms.	8	3	1

c)	With an example , describe any two types of looping constructs available in Verilog.	5	4	3
7.a)	With neat logic circuit, explain SISO and PIPO shift register.	7	2	1
b)	Differentiate asynchronous and synchronous counters. Design 2 bit asynchronous UP counter.	7	3	2
c)	Write a Verilog code for 4 bit counter in behavioral description.	6	4	3
	OR			
8.a)	Design 2 bit synchronous down counter. Write the relevant truth table and wave forms.	10	2	3
b)	With neat block diagram, describe the operation of universal shift register. Write the corresponding truth table.	10	3	2
9.a)	Describe the following terms. i) Fan in ii) Fan out iii) noise margin iv) Propagation delay	8	3	1
b)	Implement the following boolean expressions using PROM $F1(x2,x1,x0) = \Sigma m\ (0,1,7)$ $F2(x2,x1,x0) = \Sigma m\ (1,3,6)$ $F3(x2,x1,x0) = \Sigma m\ (1,4,5,6)$	8	2	3
c)	Write the neat structure of 2 input TTL NAND gate and signify any two features of the same.	4	3	2
	OR			
10.a)	Report the working of CMOS inverter with neat circuit diagram.	6	3	1
b)	Summarize the generalized structure of programmable logic devices.	6	2	2
c)	Implement the following boolean expressions using 4 input 3 output PAL(programmable Array Logic) $F1(A,B,C) = \Sigma m(1,2,4,5,7)$ $F2(A,B,C) = \Sigma m(0,1,3,5,7)$	8	3	3

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